

AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF ENGINEERING, COIMBATORE, 641112



BONAFIDE CERTIFICATE

This is to certify that the Open Lab project report entitled **"GSM BASED HEALTH MONITORING SYSTEM"**.

submitted by

CB.EN.U4ECE19249 SANJAI P K

CB.EN.U4ECE10253 SRIPATHY P

CB.EN.U4ECE10256 SANTHEEP V

CB.EN.U4ECE10259 VISHNU SANKAR P

in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology** in **ELECTRONICS AND COMMUNICATION ENGINEERING** is a Bonafide record of the work carried out at Amrita School of Engineering, Coimbatore.

Chairperson ECE
Dr. M. Jayakumar

The project was evaluated by us on:

Open Lab Coordinator
Name
Designation

Open Lab Coordinator
Name
Designation

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ABSTRACT:

GSM based Patient Health Monitoring Project mainly works in allowing doctors or relatives of patient to check the status of patient health remotely and quickly. The system comprises of Temperature Sensor, ECG and Pulse Sensor, each of which performs its assigned task that includes the calculations of heartbeat, body temperature. An immediate informative alert message will be sent to the registered number if the calculated temperature or pulse goes above a certain limit. This system is supported by Arduino UNO which is interfaced with OLED display helping the system with ECG and LCD display. The GSM based Patient health monitoring system works with GSM modem to send the data remotely to the registered number. The system also provides the privilege to the user by storing their previous checkup data in the cloud. This database will be so much of help to the doctors. The system can still be improvised and made secure with the presence of RFID scanner, increasing authenticity. The system efficiently updates doctor about health of patient as well as accurately calculates the health parameter of patient.

PROBLEM STATEMENT:

This design identifies and rectifies the delay in the transmission of the status of the health in spite of the distance and time factor.

INTRODUCTION:

The prime reason of death for men and women in many countries is heart diseases. Among them, heart attack, coronary heart disease, congestive heart failure, and congenital heart disease are the vital one, especially the countries with a very developed hospital system. The elderly persons who are in need to be monitored 24 hours a day are very much affected by the heart disease problems. Most of the time, they live alone and nobody wants to take care of them continuously. Moreover, the healthcare providing services of remote areas of many countries have not destined to an appreciate level. Considering all these circumstances in mind, a mobile monitoring system can be designed and implemented to accomplish the transmission of temperature and heart rate signal of a patient which employs Short Message Service accompanied by low-cost hardware equipment. The special feature of this system is that it can display the measured value in LCD display besides measuring the heart rate and body temperature. Hence, a patient can easily measure his / her Heart Rate and Body Temperature by himself / herself. Here, a Heart Beat Sensor Circuit is used to sense Heart Beat and Temperature Sensor to detect the temperature of a human body. Heart Beat and Temperature Sensor are interfaced. A pressure wave gets out along the arteries with a velocity of few meters per seconds when the heart beats which is faster than the actual flow of blood. One can feel the pressure wave at the wrist. But the pressure wave can rise up the volume of blood in the tissues. To detect the change, this sensor is used. Regarding the temperature, Body temperature can differ among individuals and slight variations can depend on the time of the day. This change can also depend on the activities performed by the human body and hormones generated. Despite such variations, human body has a mean temperature. The mean temperature is 98.6°F or 37°C with a variation of 1°F or 0.6°C. This value is still taken as the mean human body temperature and the set point to measure variations. The medical implications of body temperature are profound. It is the foremost medical check to confirm the health status of a human body. The change in body temperature can be used as an indicator of various diseases and medical abnormalities, including hypothermia or heatstroke, both of which can be life threatening. Body temperature can also be used as an indication of infection or other diseases and also as a means of measuring the effectiveness of fever reducing medicine. Body temperature varies depending on the part of the body from which it is measured. However, it has become standard medical procedure to check body temperature from the skin under the armpit or from the mouth or rectum. The temperature measured from the skin is understandably slightly lower and less accurate than that measured from the cavities. These sensors will collect the desired signals from the system to process them for getting the output. Then, it will feed the output signals to a transmitter. A GSM module is used to transmit the measured value from the microcontroller to the receiver unit i.e., the medical expert in forms of SMS. If we consider the terms of cost, settings and time, this system will be the most effective and profitable one than any other system currently available in all over the world. Also, this system only requires minimum man power. Besides, this system is very much helpful for those patients who live alone.

Remote Monitoring:

Remote monitoring which can also be named as self-monitoring or testing, gives the opportunity to the medical professionals to monitor a patient from a remote area by using several technological equipment. These services can provide comparatively a better health service within a very short time and in an effectively low-cost too. So, remote monitoring system has no alternative to monitor the essential parameters of a patient 24 hours a day such as heart beat, temperature etc. The system has a sensor to sense these parameters continuously and a transmitter to send a signal to the expert when those parameters have an abnormal consequence.

DESIGN AND IMPLEMENTATION:

Components and Software Used:

EDA tools required:

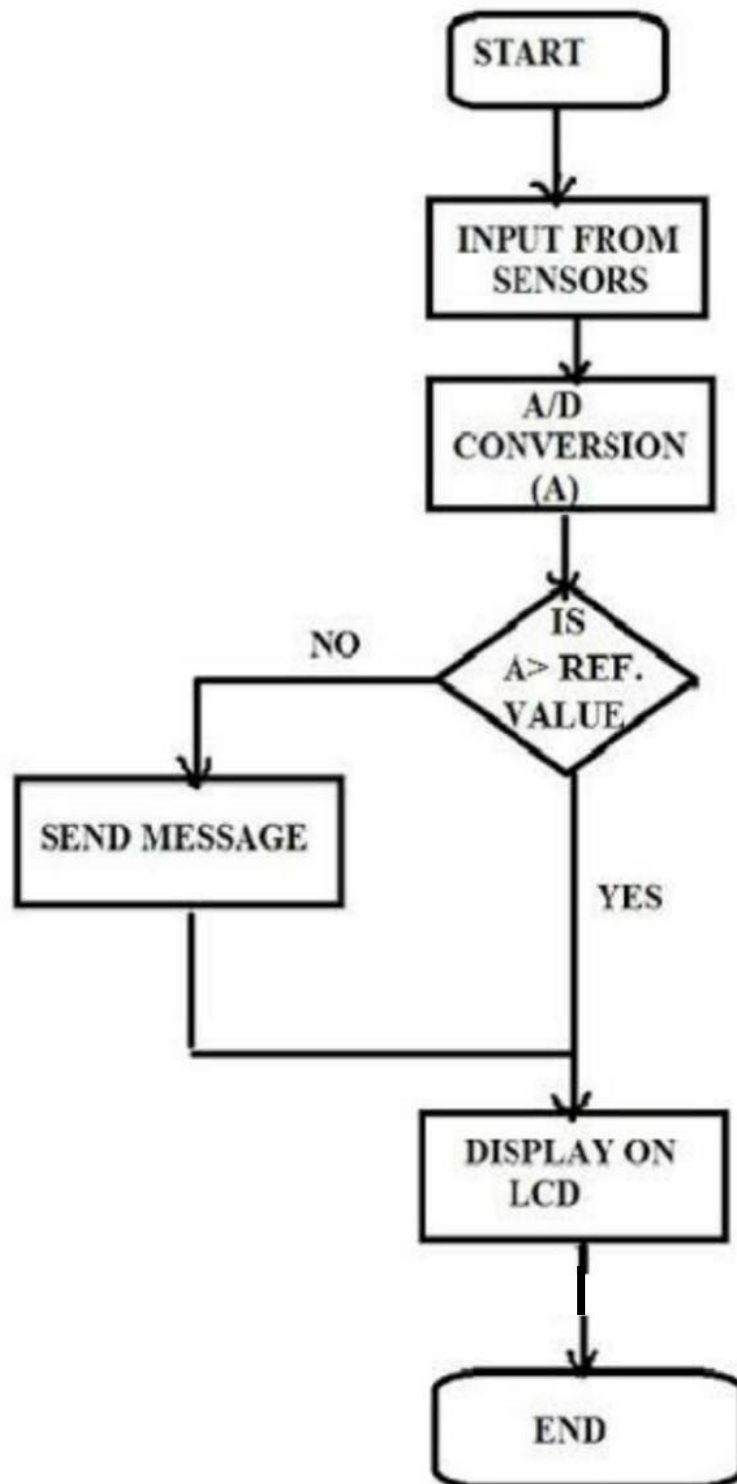
Arduino Compiler

MC Programming Language: C++

Stating work modules:

- Arduino UNO
- LCD Display
- Pulse Sensor
- Temperature Sensor
- GSM Modem
- ECG
- OLED

Flowchart:



Sensors:

Pulse Sensor:

The MAX30100 is an integrated pulse oximetry and heartrate monitor sensor solution. It combines two LEDs, a photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals. The MAX30100 operates from 1.8V and 3.3V power supplies and can be powered down through software with negligible standby current, permitting the power supply to remain connected at all times.

Applications

- Wearable Devices
- Fitness Assistant Devices
- Medical Monitoring Devices

Temperature Sensor:

The DS18B20 is a 1-wire programmable Temperature sensor from maxim integrated. It is widely used to measure temperature in hard environments like in chemical solutions, mines or soil etc. The construction of the sensor is rugged and also can be purchased with a waterproof option making the mounting process easy. It can measure a wide range of temperature from -55°C to +125° with a decent accuracy of $\pm 5^{\circ}\text{C}$. Each sensor has a unique address and requires only one pin of the MCU to transfer data so it is a very good choice for measuring temperature at multiple points without compromising much of your digital pins on the microcontroller.

Applications:

- Measuring temperature at hard environments
- Liquid temperature measurement
- Applications where temperature has to be measured at multiple points

ECG sensor:

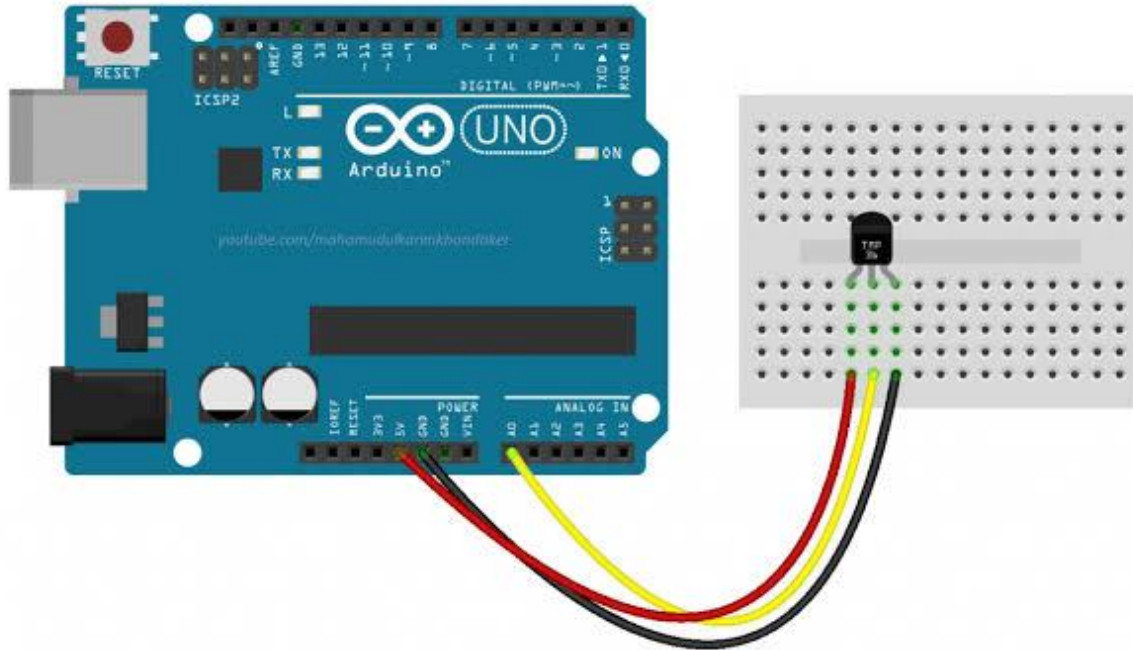
The AD8232 is a neat little chip used to measure the electrical activity of the heart. This electrical activity can be charted as an ECG or Electrocardiogram. Electrocardiography is used to help diagnose various heart conditions.

Applications of AD8232 ECG Sensor:

- Monitoring of heart and fitness activity
- Handy ECG
- Monitoring of remote health
- Used in gaming devices
- Acquisition of biopotential signal
- Biometrics
- Physiology studies
- Prototyping of biomedical instruments
- Variability of heart rate
- Interaction of human-computer
- Psychophysiology

Pin Configuration:

Temperature Sensor:

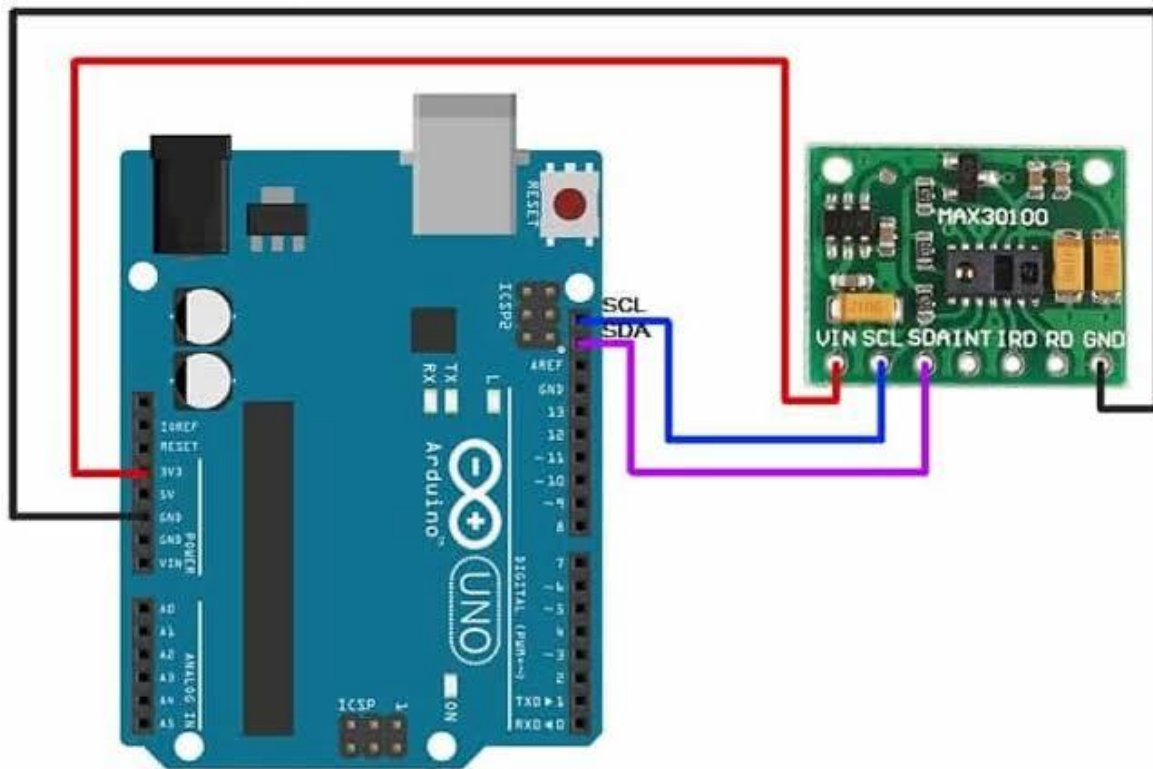


Pin1 (Ground): This pin is used to connect to the GND terminal of the circuit.

Pin2 (Vcc): This pin is used to give the power to the sensor which ranges from 3.3V or 5V

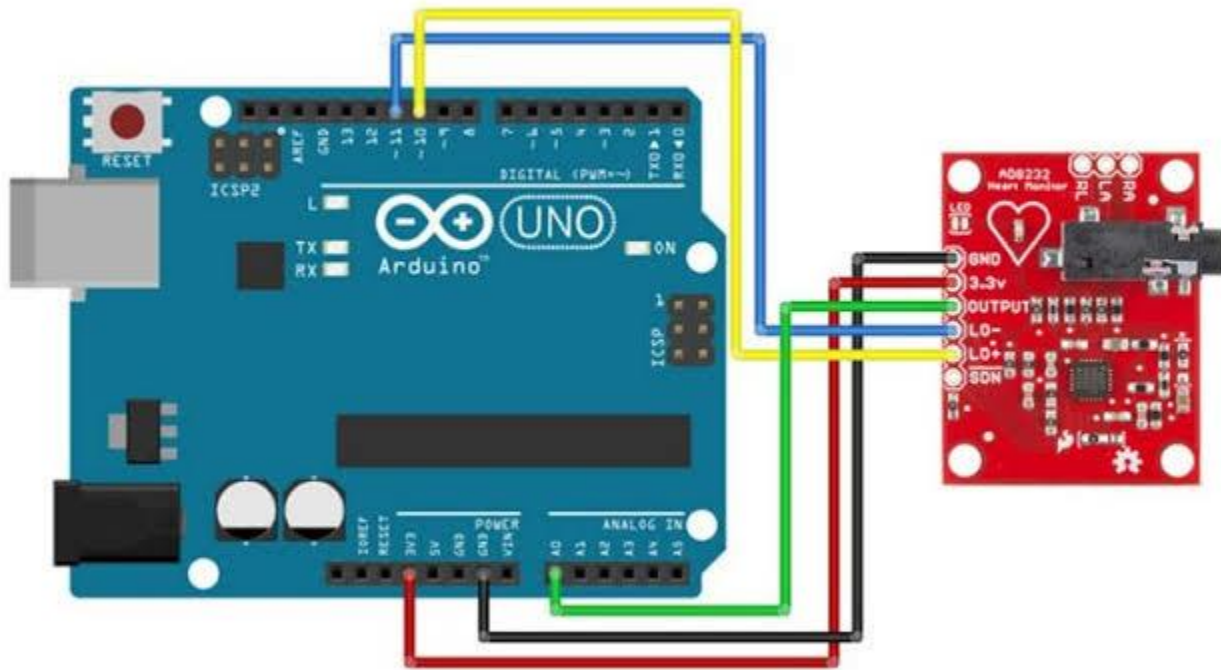
Pin3 (Data): The data pin supplies the temperature value, which can communicate with the help of 1-wire method.

Pulse Sensor:



Vin is connected to the 3.3V port of Arduino since the operating voltage of the module is 1.8V-3.3V. The ground terminal of the oximeter is connected to the ground of the Arduino. As part of the communication protocol, two pins which are the SCL and SDA are connected to the Arduino's A5 and A4 pins, respectively. The INT pin on the module is also connected to the microcontroller's Digital Pin 2 to check if the heartbeat is captured properly.

ECG Sensor:

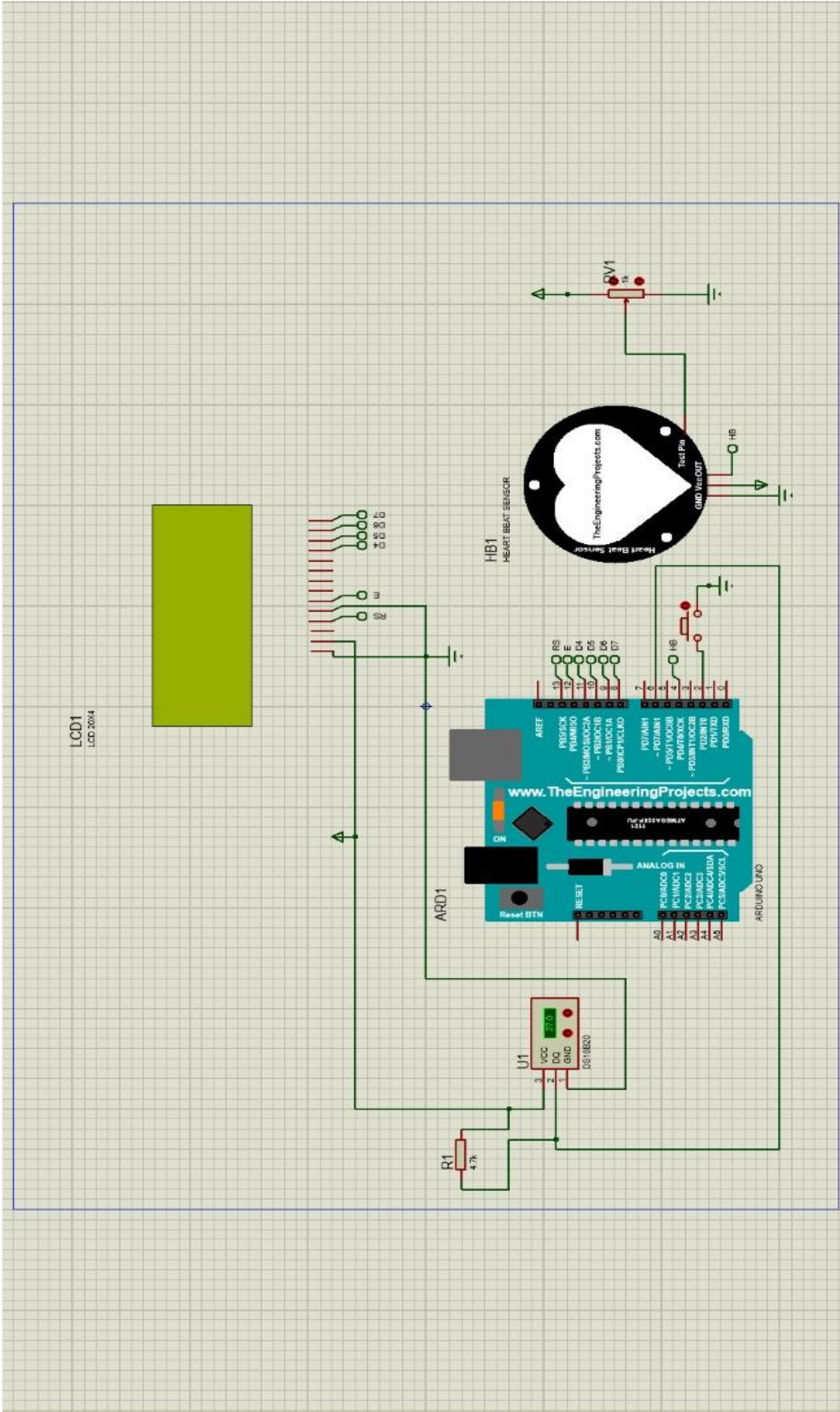


LO-: Leads Off Comparator Output. In dc leads off detection mode, LO- is high when the electrode to -IN is disconnected, and it is low when connected

LO+: Leads Off Comparator Output. In dc leads off detection mode, LOD+ is high when the +IN electrode is disconnected, and it is low when connected

SDN Pin: Shutdown Control Input. Drive SDN low to enter the low power shutdown mode.

Simulation:



WORKING:

The system comprises of three sensors interfaced with Arduino UNO. All these three sensors were picked as they give us comprehended analysis on the human health. Max 30010 is the pulse sensor employed here. The working principle of the pulse sensor is extremely simple. This sensor includes two faces where the first face is connected with an LED including an ambient light sensor whereas another face is connected with circuitry. This circuit aids in noise cancellation as well as amplification. On the front side, the LED is connected to a vein of a human body (ear tips or Fingertip), however, it should be located directly on top of a vein. The LED produces light that will drop directly on the vein. The veins in the body will have a flow of blood within them simply once the heart is pumping. So, if we check the blood flow, we can check the heartbeats also. If the blood flow is noticed then the sensor like ambient light will receive more light as they will be reflected through the blood, this small modification within received light can be analysed on time to decide our heartbeats. The electrocardiography or ECG is a technique for gathering electrical signals which are generated from the human heart. When someone experiences physiological arousal then the ECG sensor allows us to recognize the level, however, it is also used for understanding the psychological state of humans. So, an AD8232 sensor is used to calculate the electrical activity of the heart. This is a small chip and the electrical action of this can be charted like an ECG (Electrocardiogram). Electrocardiography can be used to help in diagnosing different conditions of the heart. ECG sensor is a commercial board used to calculate the electrical movement of the human heart. This action can be chart like an Electrocardiogram and the output of this is an analog reading. Electrocardiograms can be very noisy, so to reduce the noise the AD8232 chip can be used. The basic principle of working of the temperature sensors is the voltage across the diode terminals. If the voltage increases, the temperature also rises, followed by a voltage drop between the transistor terminals of base and emitter in a diode. The Temperature Sensor LM35 series are precision integrated-circuit temperature devices with an output voltage linearly proportional to the Centigrade temperature. The temperature sensor in Arduino converts the surrounding temperature to voltage. It further converts the voltage to Celsius, Celsius to Fahrenheit, and prints the Fahrenheit temperature on the LCD screen. GSM is an international standard for mobile telephones. It is an acronym that stands for Global System for Mobile Communications. Among other things, GSM supports outgoing and incoming voice calls, Simple Message System (SMS or text messaging

FURTHER DEVELOPMENTS:

The system can be further developed by interfacing it with cloud. By doing so, we could maintain a database for the patient and his medical history. This feature will be helpful for both the patient as well as the doctor diagnosing the patient. In order to increase the security of the system, RFID protocol can be employed. This ensures safety and authenticity of details of the patients who are getting tested. This entire system can be mounted on a box with two openings for Temperature and Pulse sensor respectively. With OLED and LCD forming the two graphic output structures, they can be fabricated in such a way that the graphs and the values are visible from the transparent box, the system is mounted on. With such a setup, the mobility factors sky rockets, allowing people to access the use of the system in a very easy and non-complex way. The data can be stored in cloud, for every person who take up the diagnosis. The system fitting inside a small suitcase is another added advantage of the system, making it very light and efficient to carry around. The cloud folder can be authenticated only by the medical advisor or the patient himself or herself. With this database in hand, one's medical history and vulnerability can be assessed without any delay or difficulty. Such system can be implemented in various medical stores or hospitals, to make the term of "Routine Medical Checkup", normalized. This can further grow into a regular thing, nudging people in a good way to make them aware of their health, in day-to-day basis.

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